

Can an Unsupervised Temporal Analysis Provide Useful Magic: The Gathering Data

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Abstract—In Magic: The Gathering the key to a good analysis of the game is the constant analysis of tournament winning results. These results feature human labelled categories and may fall victim to human labelling bias. Within the scope of this project the aim is to tackle a temporal analysis of Magic: The Gathering Standard format, through the lens of an unsupervised community detection algorithm. The results show that this approach is capable of detecting strong, clear and meaningful partitions, that can further be used to analyse the health of the Magic: The Gathering game without the use of human labels.

Index Terms—Magic: The Gathering, Networks, Community Detection, Louvain, Consensus Clustering, Temporal Analysis, PageRank

I. INTRODUCTION

IN collectable card games (CCGs), the aim is not only for one player to beat another whilst playing the game, but also to craft their own deck. This means adding Cards while following the rules of a given card game. In providing this experience to the player, CCGs offer two things;

- 1) An enormous search space for deck building.
- 2) A partially observable stochastic playing environment.

CCGs also pose a challenge to designers through being able to carefully design and balance an environment that can allow for experimentation, enjoyment and competition for players. Through these aspects CCGs can offer;

- 1) Mass tracking of data over time, (such as the performance of cards).
- 2) Testing of cards in a simulated environment.

These experiences offered by CCGs are great avenues for research as the environment for playing these games is complex, allowing for many approaches for game playing agents to be developed and, the vast possibilities for deck building and tracking creates an expansive search space.

In this project, the aim is to see the extent to which tracked data over time for the card game Magic: The Gathering (MTG), can lend itself to help the design decisions of the future.

A. Introduction to Magic: The Gathering

MTG, released 5th August 1993, is a multiplayer, CCG designed to be played by two or more players. It involves building a deck, using the available card pool. When playing the game players each take turns during which they play cards, with the ultimate goal to bring your opponent's life total to 0. The Game has multiple different formats (different rule sets) such as Commander (a 100-card deck format), Draft (a format where decks are made by opening packs of cards) and the

focus of this project; Standard (a 60-card deck format limited to the last 3 years of released cards (this is also known as set rotation)).

What will be studied over the course of this project is the MTG Standard meta-game, (the top performing decks in a competitive environment), over various time periods. These meta-games can be changed over time due to game altering events, these events typically include new set releases (a large batch of new cards are introduced to the game), Set Rotation (cards older than 3 years are removed from the available pool) and Banlists (specific cards younger than 3 years are removed from the available pool early). These major events and their implementations have immediate ramifications for the formats they are deployed into and can act as clear indicators of change within these formats.

Decks themselves are constructed of cards where depending on the selected format the number of unique cards has a different maximum, typically either 4 or 1. Standard has a maximum of 4 copies per deck. The following list are the main MTG card types;

- Land - a resource card type found in all decks.
- Creature - a card type that remains in the play space and is used to reduce your opponent's life total.
- Enchantment/Artifact - a card type that remains in the play space which offers benefits to the player who owns it.
- Sorcery/Instant - a one time use card type that provides immediate benefit to the player upon use.

These cards may range from colourless to one of the games 5 Main colours, (White, Black, Red, Green, Blue), or any combination of said colours. This is what creates the game's archetypes which are different play-styles brought forward by the colour system allowing decks to have unique identities and goals (one deck may seek to gain life, one may try to win as fast as possible, etc.). The resource generated by the land card type is called Mana and is spent by the player to play cards of other types. These costs are marked on each respective card and are core identifying traits of the card. It is these costs that determine the colour of a card, as Mana can be divided into the 5 colours or any combination of them, and colourless.

Finally whilst most cards are limited to decks by colour or limited to decks by play style similarities, meta staples are cards which find common inclusion due to their versatility.

B. Introduction to Project Idea

This project aims to track MTG data over a number of these time-frames, each separated by the aforementioned major

events. In doing so specific traits of these communities can be analysed so that the health and diversity of a specific time frame can be analysed. Therefore the main research questions that will be answered through this project are;

- 1) Is an unlabelled clustering approach a viable way of assessing the current state of a format and by extension a viable way of performing a temporal analysis?
- 2) Can the data provided by a temporal analysis this way provide feedback to designers free from human bias?
- 3) Can this Framework be applied to other formats and CCGs?

While many websites already do provide an analysis of the MTG standard format, from the research done on this topic none appear to analyse this data based on an unsupervised clustering method. This project will focus on that unsupervised clustering approach through the use of the Louvain algorithm.

C. Structure of this Paper

Section II, first covers all the papers that cover topics used within this paper, then will speak on the most important papers and their uses. Section III will showcase the datasets used in this project's analysis. Section IV will address the potential issues that may arise through this project. Section V will outline the methodology followed. Section VI will detail the experiments done to achieve stable graph partitions. Section VII will showcase the results, (Graphs and Tables), gathered through this project. Section VIII will discuss these results. Finally section IX will conclude.

II. LITERATURE REVIEW

A. Key findings

1) Group 1: Focuses on MTG as a research environment.

- Yannakakis and Togelius [1] was used in understanding the domain, it provided insights into AI within CCGs and acted as a source of information for other relevant papers, it broadly discussed the different types of AI implementation within card games, highlighting their potential difficulties and factors that make them an interesting research space, from the vast search space, to the complexity of playing the games themselves. The number of choices presented to the player at any given time is vast, and designing tools to navigate these environments will prove a challenge.
- Paper by Alvin et al. [2] discusses MTG as a game and further goes into detail on the structure of the game at large. It details the goals of the game, and the different types of decks encountered within it. "The game of Magic uses five colours: Black (B), Blue (U), Green (G), Red (R), and White (W)", these being defining aspects of cards which can shape any given card's identity. This is useful information as colour identity and Mana cost can determine how strong a specific card is within a specific deck. While this paper does not delve into the creation of a full agent for playing MTG, it does highlight the complexity found within these agents. Specifically, this paper asks two questions;

- Given a current hand should an agent Mulligan? (redraw the opening set of cards).
- Which card should be played for each turn?

In these two aspects of gameplay alone, the authors show the difficulty of MTG as a research area.

2) Group 2: Focuses on network analysis and community detection.

- Clustering into the correct partitions can be a very daunting task given that many clustering algorithms are not deterministic. There are many algorithms to choose and many approaches to take when it comes to a network analysis, the work by Fortunato and Hric [3], was a major part in helping narrow down the approach of this project and helped solidify a direction for it. This paper discusses many aspects of community detection and some sections in particular were taken to be used within this project. They highlight the many styles of graphical analysis and the different approaches to graphical analysis within these various styles. For example they detail two ways to approach and analyse a temporal graph. This paper proved vital in backing up the approach undertaken and it's specific contributions will be broken down in the next section.
- When determining the correct partitions for a network, a target is needed to determine which ones are better. Modularity acts as this target and scores partitions. The work by Newman [4] shows modularity as a measure of the strength of a community structure, comparing the actual observed edges of a community to that of a null model, (A null model being a random graph that preserves the degree distribution of the original graph). A higher modularity leading to stronger detected communities. However aiming for pure modularity maximization may cause issues. Shown by Fortunato and Barthélémy [5] modularity maximisation methods may suffer from a "resolution limit". This issue is one where smaller communities may inadvertently merged with larger neighbouring communities, in the aim of maximising the modularity score.
- There are many different community clustering algorithms. The Louvain algorithm offers a well established, widely used and computationally effective method. The paper by Blondel et al. [6] details the process of how the Louvain algorithm functions. The algorithm is divided into two phases, with one pass being a single run of both phases;
 - 1) A network of N nodes is started with. Each node is it's own community. For each community i evaluate modularity gain from joining with each neighbouring community j . Make the move with the highest modularity yield. If no positive modularity gain can be made for i , i remains its own community. Iterate until no improvements to modularity can be made.
 - 2) A new network is built, whose nodes are the communities found in phase 1. Weights of edges, are the sum of the weights of the edges between communities in phase 1. Edges between nodes in

the same communities become self-loops.

Each pass reduces the number of communities, and starts with either the initial network or the one formed in phase 2. This is iterated until maximum modularity is achieved. An important aspect of the Louvain algorithm is that it is stochastic and its output can depend on the initial choice of node.

- As stated earlier most clustering algorithms are not deterministic. The Louvain algorithm for example can have many different results when running on the same graph. The work by Lancichinetti and Fortunato [7] shows that consensus clustering when combined with an existing algorithm can produce partitions with increased stability and accuracy. They detail a framework which is not a modularity maximisation technique instead it iteratively applies an algorithm to produce better partitions than that of the algorithm by itself. The full consensus clustering workflow is as follows;

- 1) Apply the selected algorithm on selected graph n times to produce n partitions.
- 2) Compute a consensus matrix D , where D_{ij} is the number of partitions where nodes i and j were assigned to the same cluster, divided by the number of partitions n .
- 3) All entries of D that are below a selected threshold τ are then set to 0.
- 4) Form a new graph using D .
- 5) Apply the selected algorithm on new graph n times.
- 6) if all partitions are equal, stop. If not repeat process from step 2.

- The question of the correct partition within community detection is still a difficult question to answer. So a way to compare to the ground truth is important. The work by Danon et al. [8] suggests the use of Normalized Mutual Information (NMI) to compare to the ground truth. NMI involves building a matrix of real labels verses found labels. In each cell N_{ij} is a count of how many nodes are in community i and ended up in community j . Using this they can tell that if all the nodes of a label ended up within the same community, and they are the only nodes in said community, the two line up perfectly. If the found partitions are identical to the real communities the final NMI score will be 1. This measure of accuracy can provide a performance metric to the algorithms employed within this project.
- Adjusted Rand Index (ARI) is another method that will be used to judge partitions. The work by Hubert and Arabie [9] goes through explaining it. ARI compares two sets of labels (U and V) for a set S , by comparing elements pairwise. For each pair of elements in S there are 4 outcomes;
 - 1) Objects are in the same class in U and the same class in V .
 - 2) Objects are in different classes in U and different classes in V .
 - 3) Objects are in the same class in U and different classes in V .

- 4) Objects are in different classes in U and the same class in V .

Naturally from this a number of agreements and a number of disagreements. Using these agreements the similarity between two partitions can be determined. The specific addition this paper added to the Rand Index, was to correct for the score, added by the number of agreements that would appear simply due to chance.

- Raw community counts detected by the previous algorithms alone may prove to be disingenuous. It is important to be able to distinguish how much information they carry and it is just as important to be able to determine the balance communities have within their respective graphs. Using Shannon entropy from the work by Shannon [10], the measure of a partitions balance and evenness can be determined. Shannon entropy alone can't be compared across snapshots as the maximum value for Shannon is based upon the number of communities. By normalising the Shannon entropy number by the log of the number of communities Pielou's evenness index is obtained. The work of Jost [11] highlights the use of Pielou's evenness index as a competent measure of evenness. Which can act as a clear singular value indicating the balance of the current communities.
- PageRank is another measure used within this project. Based on the work by Brin and Page [12], PageRank itself is based off of a random walk, however it has a random chance to teleport to another node instead of following an edge, to avoid the common pitfalls of a random walk. It works off of the premise that important nodes are pointed to by other important nodes. This PageRank value then provides an overall importance score for each node within the graph.

B. Discussion of Papers

This section will detail the most important contributions of the following papers, found to be important for this project;

1) *Community detection guide - Fortunato and Hric [3]*: as previously mentioned added a lot to this project. The first introduced the idea of a temporal analysis which involves analysing a series of static graphs over time, "Typically one can describe it as a succession of snapshots $G_1, G_2, \dots, G_L$, where each snapshot G_t corresponds to the configuration of the graph in a given time window".

Using the major events in MTG as separation events, the use of a succession of snapshots for analysis is integrated seamlessly. Tournament results from each time-frame can be converted into a snapshot resulting in a graph and further information for analysis.

The next useful benefit it provided was highlighting Jaccard Similarity as a method to compare communities or groups of nodes to one another. Here Jaccard similarity works by taking the count of the intersection of two sets and dividing it by the count of the union of two sets. This can provide a useful measure of similarity which was used throughout the project.

Finally this paper highlighted consensus clustering as a method in which to combat the random nature of some clustering algorithms.

2) *Louvain Algorithm - Blondel et al. [6]*: provides a strong well established, widely used and computationally effective unsupervised community clustering algorithm, which will form the basis for this project. The algorithm has many advantages such as it's computationally fast speed. Louvain is a modularity maximisation technique so as spoken about earlier with the works of Newman [4], and Fortunato and Barthélémy [5] it can fall victim to the resolution limit, this is a limitation that may affect this project and is worth paying attention too.

3) *Consensus Clustering - Lancichinetti and Fortunato [7]*: consensus clustering provided much to this project. The researchers stressed within the paper that the goal of consensus clustering was not to find a better optimum for algorithms, (e.g. will not try to find a better modularity for Louvain), instead it aims to reduce the stochastic nature of these algorithms. Using the workflow from the previous section consensus clustering iteratively uses the selected algorithm to find partitions. This is all in effort to produce a consensus matrix whose connections between communities are weak and connections within them are strong. Once a consensus has been reached by consensus clustering a final pass of the selected algorithm can be used to produce the final partitions for analysis.

In effect consensus clustering seeks to reduce the stochastic nature of clustering algorithms by running them numerous times and aggregating information to produce more stable partitions.

This provides benefit to this project through making produced partitions repeatable and consistent without having to rely on setting a random seed.

As the paper included the Louvain algorithm as one of its tests it was able to show how the algorithm improved through the employment of consensus clustering. Specifically they show how it improved in terms of stability.

The researchers also found that employing the use of consensus clustering was able to find better partitions than simply using the selected algorithm alone, within their testing. Providing more evidence as to why consensus clustering will prove useful for this project.

Important to note is consensus clustering does not remove the limitations of community detection, nor does it seek to optimize modularity, it simply seeks to provide stability.

The team also did employ consensus clustering on a temporal network, however their approach differs quite a bit from the succession of snapshots method undertaken within this project.

Based on the previous workflow of the prior section, consensus clustering has two main inputs, n (the number of times the selected algorithm is used within each iteration), and τ (the threshold value). And it is these two values which will have to be experimented on to receive good results. The team do mention that Louvain performs well with a lower threshold which offers a good starting point for experimentation. They also state that increasing the number of runs of the selected

algorithm may not increase performance, so this also provides another useful starting point for experimentation.

4) *NMI - Danon et al. [8]*: NMI provides a useful measure for this project as it can allow for an overall confidence score for how strong a particular partition is compared to the ground truth. Essentially it can help determine how much information is in one of the detected communities.

As NMI allows for comparison between two labels substituting out the ground truth above for another detected partition can help determine the similarity between two partitions. Using this it can be determined if a method is stable by using different runs and seeing how different its partitions are. NMI will be used throughout the project as both a measure of stability and a comparison to the ground truth.

5) *ARI - Hubert and Arabie [9]*: ARI provides another useful measure for this project as it can show how consistently the found communities reflect the groupings of the ground truth adjusted for chance. It should be noted that if the number of found communities is far lower than the number of labelled archetypes ARI will naturally fall, however it's this information in conjunction with the other statistics that will be gathered, that will provide an detailed overview of the detected communities.

6) *Shannon entropy and Pielou's evenness index - Shannon [10] and Jost [11]*: Shannon entropy can be used to determine the balance of communities across a graph this in combination with a detected number of communities can help determine the true diversity of a graph. If one community represents 90% of individuals the graph is not as initially diverse as suggested. This metric can help track diversity for each produced graph. While Shannon entropy alone can not be used to compare different graphs when normalised into Pielou's evenness index it can act as a competent comparative value across snapshots.

7) *PageRank - Brin and Page [12]*: PageRank allows another method of tracking data, here the PageRank scores for nodes can be obtained in various graphs showcasing how important that node is within this graph. This can lend itself to a temporal analysis as nodes can be deemed as important and tracked to see if they remain important.

III. DATASETS

The main data set used for this project was MTGTop8 [13]. This is a website that consists of tournaments for MTG and details the deck lists for each tournament. Numerous tournaments are hosted throughout different points of the year and tournaments are rated by competitiveness (1 small star, 2 small stars, 3 small stars and finally 1 large star). As CCGs have large events that disrupt the flow of a meta-game, (Banlists, new set releases and Set Rotations), it was determined that various snapshots would be acquired each representing a different period in the standard meta-game, whilst being separated and distinguished by the occurrence of one or more of the aforementioned major events. The following 8 snapshots were taken for this projects analysis;

In Table I the names, (not including Set Rotation and Banlist), refer to set names. The Start dates and End dates are not the specific dates in which in the major events occurred

Snapshot	Start event	Start date	End event	End date
1	Aetherdrift	16-02-25	Tarkir: Drag-onstorm	10-04-25
2	Tarkir: Drag-onstorm	19-04-25	Final Fantasy	12-06-25
3	Final Fantasy	14-06-25	Banlist	29-06-25
4	Banlist	01-07-25	Set Rotation + Edge of Eternities	27-07-25
5	Set Rotation + Edge of eternities	09-09-25	Marvel's Spider-Man	25-09-25
6	Marvel's Spider-Man	27-09-25	Banlist + Avatar	11-09-25
7	Banlist + Avatar	22-11-25	Lorwyn Eclipsed	22-01-26
8	Lorwyn Eclipsed	24-01-26	TMT	05-03-26

TABLE I
SNAPSHOT DATASET INFORMATION

instead they are surrounding days, which allowed for results to be restricted within these time frames.

To ensure the finalized graphs that were analysed held strong data about each community, only results from tournaments rated 2 stars or higher were taken and furthermore only the top 8 decks from each of those tournaments was taken. On the rare occasions that a tournament did not have a registered deck list for a top 8 finish, less than 8 decks were taken (whichever remaining decks had full deck lists available). This was to ensure that only strong decks were considered for the analysis, as without this filter any cards could go together in any deck. For this project it is the meta-game defining structure that will be analysed. The dates above were used to constrict snapshots to time frames between major events, they usually proceed or follow major events by a couple of days, whenever tournaments start up again.

Decks may also include a side deck, this is an alternative set of cards, that a player may swap in in-between games for an advantage, (akin to substituting players in sports). While a part of the meta-game these cards are typically used to deal with specific other decks and therefore will not be added to the graph as they are not main parts of the communities that will be detected.

A secondary dataset used in this project was Scryfall [14] is an open-source dataset provided by Scryfall which contains all standard legal cards as of May 8th, 2026. It contains notable information on all cards, Relevant MTG game related information, such as Mana cost or colour identity (alongside many other card identifiers), as well as non-game play related information such as cost of cards and rarity. This dataset was used to sculpt which cards appeared on the community graph analysis. This is because land cards, may cause disconnected and unrelated communities to become one on the basis that the share a colour not because they have any meaningful data. This is not entirely true for all land cards, some land cards are dual sided (the front and back are different cards, essentially making the single card two in one). So any cards with the type line "Land" or "Land - ..." were removed and not placed on the graph however cards with type lines like "Artifact // Land"

were placed on the graph.

IV. CHALLENGES

A. Limited Data

A major challenge experience by this project is limited data. The graphs that are used within this project are taken from real life tournament results and are dependent on them. As previously mentioned in the previous section results were only taken from tournaments rated 2 stars or higher as to ensure that only meta defining decks were considered for the analysis.

Snapshots are also dependent on the various time-frames defined by the implementation of major changes within the card game.

Due to these changes occurring at various points all the snapshots have a different number of decks drawn from to formulate their results, to account for this, values were normalized to allow them to be more comparable across snapshots.

B. Fun

A CCG at its core is a game and a game's aim is to be fun. It's this measure that can present another challenge for this project. Later on diversity of a format is used as a positive measure (the number of available decks), as it allows for more creativity and play-styles to be present within a format. Some players may however disagree with this and prefer a format dominated by 1 or 2 decks, as if the tools both players have available are the same the fun may come from skill expression and who may pilot a deck better.

Cards may also be banned for being not fun to play against despite an inability to perform well in tournaments.

As fun is immeasurable and different for each player it will not be a major focus of this project. Measurements such as diversity instead will be used to measure success.

V. METHODOLOGY

A. Snapshot Selection

For the selection of snapshots (or later graphs), some already completed competitive formats from the previous year were selected ranging from 16-02-25 all the way to the 05-03-26. All the decks gathered, from any specific time frame were all collected into a singular graph representing that time frame. Due to the massive differences between snapshots, with cards leaving and joining, and clear major events distinguishing these snapshots, they are treated as separate entities, producing 8 separate graphs instead of 1 unified one over time.

B. Louvain Algorithm

For the formation of partitions the Louvain algorithm was used. To ensure a fair comparison across all snapshots the default Louvain parameters were used.

C. Consensus Clustering

As the most important part of the analysis lies within the communities found by Louvain, consistent and correct partitions have to be guaranteed.

Firstly consensus clustering using the Louvain algorithm is implemented. After ensuring correct implementation, a multitude of runs are performed to ensure stability and find good parameters for consensus clustering (number of times Louvain is done each iteration and threshold value).

A range of parameters was selected (50, 100, 200), for the number of times Louvain is ran and (0.2, 0.3, 0.4, 0.5) for threshold values. Each combination of these values was run 5 times, this was done to confirm that subsequent runs with the same parameters, could produce similar partitions, this was tested using Normalized Mutual Information (NMI), which returns a score based on how similar partitions are. Meaning that a high NMI score indicates that two partitions share a high degree of structural information.

Within these five runs for each parameter pair the final consensus clustering run would be saved. This was done so a comparison across parameter pairs could be carried out and to see if the output partition heavily relied on the consensus clustering parameters.

After this final comparison, the strongest parameters for consensus clustering can be determined so that they may be used in the temporal analysis of the graphs. This process was repeated for each Snapshot.

Finally when having achieved the optimal parameters for the final assessment, Louvain consensus clustering will be ran 10 times for each snapshot taking the majority vote as the graph that will be used for this project's analysis.

D. Graph Statistics

After securing a reliable Louvain algorithm that can produce strong partitions, each snapshot is ran through the algorithm, each producing the following:

- An undirected weighted graph.
- A directed weighted graph.
- Two dictionaries of cards and their number of occurrences, one normalized and one not.
- A dictionary of cards and their PageRank scores within the graph.
- The modularity of the graph.
- The number of communities in the graph.
- Useful graph statistics such as number of nodes, edges, density, average degree and average edge weight.
- A dictionary of communities and all the nodes in each of them.
- The top 10 cards by occurrences per community.
- The top 10 cards by PageRank per community.
- The density within each community.

The weighted graph will be weighted by normalised occurrences, the number of times cards appeared in the same deck divided by the number of decks. This is the graph that the consensus Louvain analysis will be performed on and the graph the partitions will stem from.

The directed graph will be weighted based on conditional probability, (i.e weight equals the probability that the pointed to card is in a deck given that the pointing from card is already in that deck). This graph will be used for the PageRank analysis.

Graphs can also be analysed to see if recent events added diversity or decreased diversity in a particular meta-game. Allowing for the designers of the game to receive mathematically backed feedback on how a set performed. It would also allow for the comparison of sets, seeing which ones brought up diversity at the times of their releases and which ones dropped it.

The modularity obtained from these graphs over time can also be used to perform an analysis. It can show that the communities detected are better than random. The modularity score measures the quality of the partitions where higher scores mean more clearly defined structures, showing how clear the detected communities are.

E. Detected Communities Vs. Human Labels

The current methods outlined for community detection have followed an unsupervised method (Louvain), here no labels are used in the formation of the found communities. To determine 1) if the detected communities have meaning and 2) the meaning of these communities, a comparison to the human made labels for each deck was made.

In this pipeline, firstly the number of different archetypes and their occurrences within each snapshot are gotten. It can then show how many of each archetype fall into the snapshot's communities.

Secondly the eligible cards within each deck is obtained. The eligible cards here refer to the fact that containing multiple copies of a card will count multiple times for community assignment, instead of only counting unique cards, as containing more of an archetype card would naturally push that deck to being within a specific archetype and by extension community. Land cards are not counted as eligible cards. Appearance scores, (total number of eligible cards that appeared in community x divided by total eligible cards), for each community for each deck are gotten and each deck receives a community label. Decks with appearance scores above 40% are sorted into a clean assignment category and below 40% an ambiguous category providing an overall confidence split for the snapshot.

NMI Scores can also be calculated here comparing the ground truth to the assigned communities showing how close the predictions were to the ground truth. It is how much predictive information lies within a snapshot's found communities. ARI on the other hand compares pairs of nodes across a graph and tracks whether or not those pairs are grouped the same compared to the ground truth. ARI reveals how consistently the found communities reflect the groupings of the ground truth adjusted for chance.

F. Diversity of the Snapshot

Going on from here the contents of the decks both, within each communities and across communities can be compared.

Providing each community with an intra-community similarity score and an inter-community similarity score. This will be done using Jaccard similarity. Averaging these across the snapshot can return two values which can be used in an intra and inter-community similarity comparison. If a large portion of decks within any given community are similar, and dissimilar to the decks of other communities, a good indication with evidence can be ascertained if the community found is a real cluster that holds decks that do resemble one another.

Using the number of decks per detected community the evenness of the communities can be obtained.

As previously mentioned the raw number of communities graphed over time alone may be misleading if 90% of the decks stem from a single snapshot. Representation per community per snapshot will also be printed so it can be analysed. As well Shannon Diversity Index value can be calculated and normalised into a Pielou's Evenness Index which can showcase how balanced the communities of a partition are.

G. New Card Integration

The integration of new cards can also be tracked. What specifically can be tracked here is how many cards that were not present in the previous snapshot entered the competitive metagame, and to which communities they joined. The sets they joined from can also be determined. Meaning a specific count of how many cards from the most recent set that saw competitive success and a percentage of the newest set saw competitive play can be obtained. Allowing for another mathematically backed measure of a set.

H. Tracking Communities Over Time

Communities from snapshots can be tracked across subsequent snapshots, seeing if they survive or not. They can be tracked using Jaccard similarity and specifically tracking the most important individuals within each community, identifying those individuals through the use of PageRank.

I. Alternative question

Despite the main research questions outlined in the introductory section of this paper. There is an alternative question that may be answered further through the use of this data.

1) Can evidence be provided for the banning of a card?:

By tracking the performance of previously banned cards can we determine what classifies as worthy for a ban for the MTG design team and then in turn provide a numerical measure that could argue for the banning of a current card.

Here, specifically a card's normalised occurrence and PageRanking for each snapshot can be tracked, studied and determined if they can be used as viable indicators of a cards need to be banned.

VI. EXPERIMENTS

The main experiments carried out for this project were the tests to get the optimal parameters for consensus clustering. As previously mentioned 50, 100 and 200 were selected as

Parameters	SS1	SS2	SS3	SS4	SS5	SS6	SS7	SS8
50, 0.2	0.974	0.997	0.980	0.976	0.978	0.946	0.948	0.950
100, 0.2	0.971	0.988	0.980	0.976	0.955	0.981	0.963	0.964
200, 0.2	0.976	0.996	0.979	0.976	0.978	0.981	0.965	0.974
50, 0.3	0.987	0.995	0.969	0.958	0.955	0.975	0.948	0.956
100, 0.3	0.996	1.0	0.969	0.972	0.978	0.956	0.961	0.964
200, 0.3	1.0	0.993	0.969	0.984	0.972	0.995	0.991	0.978
50, 0.4	0.992	0.990	0.979	0.972	0.995	0.971	0.975	0.975
100, 0.4	0.994	0.996	1.0	0.985	1.0	0.997	0.988	0.975
200, 0.4	1.0	0.996	0.979	0.985	0.993	0.995	0.993	0.975
50, 0.5	0.989	0.992	1.0	0.978	0.976	0.996	0.968	0.985
100, 0.5	0.992	0.993	1.0	0.981	0.990	0.992	0.982	0.980
200, 0.5	0.991	0.991	1.0	1.0	0.995	0.996	0.990	0.981

TABLE II

NMI RESULTS - WITHIN PARAMETER

the possible values for number of time Louvain is carried out while 0.2, 0.3, 0.4, 0.5 were used as threshold values.

Table II shows the average NMI value across the 5 runs, carried out for each version of the parameters, within each snapshot. (E.g in snapshot 1, the parameters 50 Louvain runs and 0.2 threshold had an average NMI score of 0.974).

All the values in Table II are quite high indicating that the structure detected by the algorithm was very consistent at all levels. The highest average across snapshots was 200 Louvain runs and a 0.5 threshold. For this reason these were selected for the final analysis.

To confirm the detected structure, did not heavily rely on the selected parameters, another check was performed. The final run for each parameter pairing was stored and compared across each parameter pair for each snapshot. (e.g the last run for 50, 0.2 and so on were all stored, then NMI was calculated comparing all parameter pairs against one another for each snapshot, those results are shown on the table below).

Snapshot	Average across NMI all parameters
1	0.985
2	0.995
3	0.975
4	0.963
5	0.973
6	0.977
7	0.970
8	0.970

TABLE III

NMI RESULTS - ACROSS PARAMETER

As seen in Table III the results for NMI across the different parameter values were quite high meaning that the detected structure of the graph did not rely on specific parameters for consensus clustering to achieve the structure.

VII. RESULTS

A. Graph Statistics

Despite all the experiments of the previous section designed to reduce the randomness of Louvain into more stable partitions, the volatile nature of CCGs and the general structure of their produced graphs, still had some variance. As seen by the experiments, the overall structure was determined to be quite solid however the randomness of a few nodes caused the number of detected communities within each snapshot to vary slightly. Again the majority of the graphs structure was

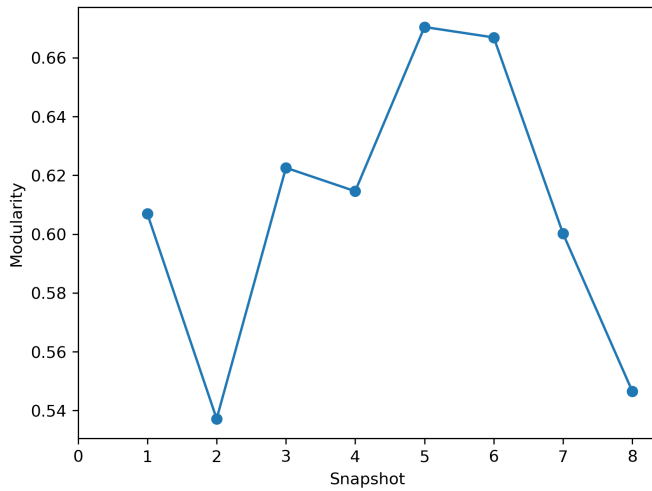


Fig. 1. Modularity of snapshot's graphs over time

consistent. To account for the small number of nodes causing these issues, the final consensus clustering Louvain algorithm was run 10 times for each snapshot's graph. The most common partition found within these 10 was used for the final analysis of the graph.

Snapshot	Decks	Nodes	Edges	Is the graph connected
1	697	403	5852	Yes
2	986	415	6539	Yes
3	210	213	2289	Yes
4	325	423	5764	Yes
5	184	211	2153	Yes
6	542	349	4236	Yes
7	889	524	8729	Yes
8	678	410	6267	Yes

TABLE IV
STATISTICS FOR EACH GRAPH

Along with the data collected from Table IV both the modularity and number of communities was collected over time, each being represented by Fig. 1 and Fig. 2 respectively.

As seen in Fig. 1 the modularity of the format fluctuates over time. The modularity score measures the quality of the partitions meaning that it can tell if the communities detected are better than random, where higher scores mean more clearly defined structures.

As the scores here range from 0.54 - 0.68 it can be said confidently that strong, well defined communities were detected throughout the results for each graph.

Fig. 2 on the other hand shows the number of detected communities over time. These can be used as a measure to determine the diversity of a time-frame within a format.

Snapshot	Number of cards lost
1	150
2	239
3	49
4	285
5	27
6	79
7	213

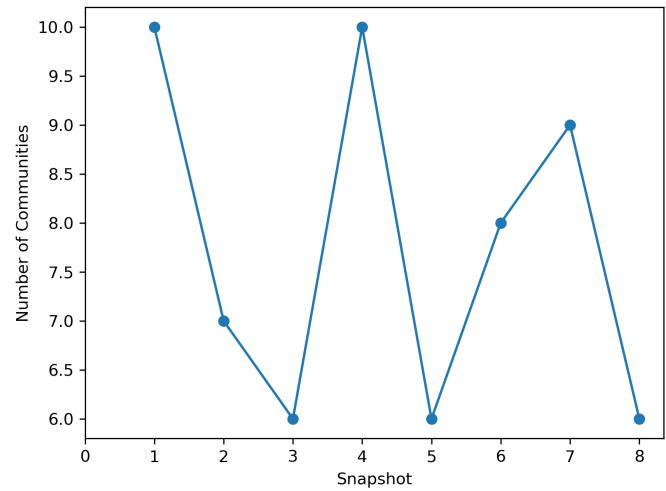
TABLE V
NUMBER OF CARDS THAT FELL OUT OF THE META

Fig. 2. Number of communities of snapshots over time

Community	SS1	SS2	SS3	SS4	SS5	SS6	SS7	SS8
0	59	89	30	41	21	44	85	90
1	54	36	34	21	25	63	45	18
2	49	20	67	105	56	32	25	85
3	60	43	33	59	54	98	107	80
4	11	91	23	11	16	1	32	92
5	11	36	26	10	39	49	73	45
6	10	100		14		29	5	
7	40			57		33	88	
8	71			71			64	
9	38			34				

TABLE VI
NUMBER OF CARDS IN EACH SNAPSHOT'S COMMUNITIES

Table V shows the number of cards that did not show up in subsequent snapshots (cards that did not achieve any tournament wins and therefore fell out of the meta)(e.g 285 cards were lost in the jump from snapshot 4 to 5).

Table VI shows for each community in the Snapshot how many unique cards were present within it, (empty space meaning that snapshot did not have that many communities). Community 4 in snapshot 6 had only a single card which is not substantial enough information to represent any decks within the metagame so that community was dropped. For the remainder of this analysis snapshot 6 will be treated as having 7 total communities. While the next lowest card count is community 6 in snapshot 7 with 5, however a deck can contain 4 copies of the same card and as this table only represents unique cards. With 20 cards, this community can represent a strategy, with the remaining slots in the deck being dedicated to lands and meta staples. For this reason and for it being only a singular occurrence within the entire analysis community 4 in snapshot 6 was dropped.

B. Comparison to User Labels

As the process so far has been unsupervised community detection, no labels have been used in the analysis. To ensure that the detected graphs were indeed identifying a meaningful structure they can be compared to human made labels of deck archetypes.

Snapshot	Number of cards	Number of Decks	Number of archetypes	Number of communities	Confidence split	NMI Score	ARI Score	Intra community similarity	Inter community similarity
1	403	697	63	9	0.997	0.669	0.404	0.664	0.311
2	415	986	59	7	0.999	0.681	0.586	0.584	0.308
3	213	210	23	6	1	0.802	0.773	0.63	0.29
4	423	325	52	10	1	0.699	0.55	0.689	0.313
5	211	184	27	6	1	0.771	0.661	0.616	0.267
6	349	542	45	7	0.996	0.778	0.717	0.672	0.286
7	524	889	80	9	0.987	0.662	0.335	0.595	0.299
8	410	678	77	6	0.99	0.652	0.374	0.556	0.322

TABLE VII
COMPARISON TO HUMAN LABELS

Com	SS1	SS2	SS3	SS4	SS5	SS6	SS7	SS8
0	194	102	5	19	25	82	90	180
1	0.28%	0.10%	0.02%	0.06%	0.14%	0.15%	0.1%	0.2%
2	34	325	27	12	55	32	44	91
3	0.05%	0.32%	0.13%	0.04%	0.3%	0.06%	0.05%	0.13%
4	27	7	29	47	22	170	18	102
5	0.04%	0.01%	0.14%	0.14%	0.12%	0.31%	0.02%	0.15%
6	130	113	75	98	30	87	231	58
7	0.19%	0.11%	0.36%	0.30%	0.16%	0.16%	0.26%	0.09%
8	39	188	7	1	2	49	137	137
9	0.06%	0.19%	0.03%	0.00%	0.01%	0.06%	0.2%	0.2%
10	7	8	67	1	50	126	117	110
11	0.01%	0.01%	0.32%	0.00%	0.27%	0.23%	0.13%	0.16%
12	1	243		1		35	1	
13	0.00%	0.25%		0.00%		0.06%	0.00%	
14	39			8		10	88	
15	0.2%			0.02%		0.01%	0.1%	
16	116			115			251	
17	0.17%			0.35%			0.28%	
18	10			23				
19	0.01%			0.07%				
Total	697	986	210	325	184	542	889	678

TABLE VIII
PERCENTAGE OF DECKS PER COMMUNITY

total decks used to build the respective snapshots graph. This count allows for a deeper dive into the diversity of each snapshot through the evenness of its communities.

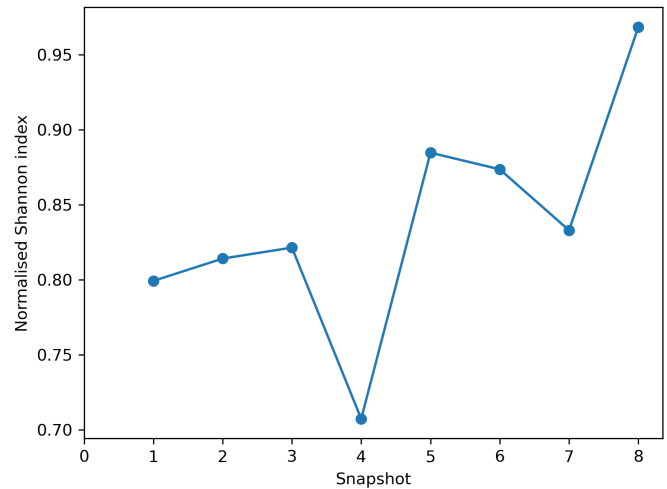


Fig. 3. Pielou's Evenness Index

Table VII shows that the number of communities detected was far less than the number of labelled archetypes present for all snapshots.

The NMI scores here, ranging from 0.652-0.802, show that within each community the archetypes present, represent a small consistent cluster of archetypes and not a range of archetypes.

The ARI scores fluctuates quite a lot within the statistics for each snapshot ranging from 0.335-0.773 however a wide range of ARI values are unavoidable due to the fact a far fewer number of communities are detected when compared to human labels.

The implications of NMI and ARI will be spoken on in further detail within the discussion section.

Finally Table VII shows that the vast majority of decks could confidently be labelled as belonging to a community shown by the confidence split.

C. Diversity of the Snapshot

The intra community similarity values, from Table VII, range from 0.556 - 0.689 whilst the inter community similarity values range from 0.267 - 0.322, these values were achieved using Jaccard similarity over the available decks in each community.

Table VIII shows per snapshot per community, both the number of decks and the percentage they represent from the

Fig. 3 shows that using the percentage of decks per community Shannon's entropy can be calculated, and then normalised into Pielou's Evenness Index as a singular value representing the evenness of a graph's communities. Which over time can be tracked, which will help track a format's diversity. This value can reveal how balanced the found communities are. The values found range from 0.7-0.97.

It is important to note that no archetype labels are needed for this as decks are placed into communities based on raw card counts and those decks per communities are what are used to get Fig. 3 and Table VIII.

D. New Card Integration

Another measure of success that can be used for specifically sets, is the integration of their new cards. Here the percentage of cards that were able to see immediate competitive success can be determined.

Firstly the sets *Aetherdrift* and *TeenageMutantNinjaTurtles* (TMT) were not a part of this analysis found in Table IX as the snapshot before *Aetherdrift* and the snapshot after TMT were not a part of this project's scope. Table IX details the total number of standard legal cards introduced with a set (Total cards), the

Set	Snapshot Introduced	Total cards	Total without lands	Tournament success
Tarkir: Dragonstorm	2	308	270	45
Final Fantasy	3	328	298	15
Edge of Eternities	5	283	257	21
Marvel's Spider-Man	6	193	178	16
Avatar	7	286	262	60
Lorwyn Eclipsed	8	283	266	38

TABLE IX
NEW CARD INTEGRATION

total number of cards that could have been added to their respective graphs (Total without lands) and the number of cards that saw immediate tournament success within the snapshot they were introduced.

E. Tracking Communities over time

The next section of this project involves tracking specific communities over time. This involved using Jaccard similarity to determine which community the current one evolved into over the jump into a new snapshot, or seeing if said community dissolved.

Table X showcases the Jaccard similarity between Snapshots 1 and 2. The scores here are quite low throughout, but because cards will enter the meta game (both new and already existing) and some cards will fall out of the meta game this table alone can not say whether or not a community survived.

The findings of any Jaccard Table X are backed up using PageRank. Specifically the highest rated individuals with each community are treated as the most important individuals, so tracking these over time alongside Jaccard similarity can help determine what happens to the communities. The last two columns represent the number of cards that did enter the next snapshot (snapshot 2 here) and the number of cards that did not enter that snapshot.

New cards can also be tracked, to see how many cards entered the meta game and to which community they joined.

Table XI showcases the number of cards that were not present in the previous snapshot, that saw competitive play in the current snapshot and to which community they joined. Note it is possible for a card to join the metagame, leave and rejoin at a later point. This table simply counts cards not seen in the previous snapshot.

So to run an example, through the use of Table X it shows the majority of snapshot 1 community 0's (1-0) cards where split between snapshot 2 community 1 and 3 (2-1 and 2-3 respectively). Through specifically tracking 1-0's most important cards by page rank it can be shown 7 of them landed in 2-3 with 16 cards not seen in snapshot 1 being introduced and joining this same community shown in Table XI.

As the most important cards survived and stuck together, and within snapshot 2 those same cards remained the most important cards, shown in Table XII it can be safely said that 1-0 cleanly survived into snapshot 2 with some new cards joining onto it.

Table X Can also determine if an archetype dies out, shown by snapshot 1 community 5, here 9 out of its 11 cards fall out of the meta and see no tournament wins in snapshot 2. Here it

can confidently be said that this community died out while the few remaining cards were absorbed into another community.

Table XII showcases the top cards by overall graph PageRank (Both card name and PageRank score), in snapshot 1 Community 0 and snapshot 2 Community 3 to showcase how the important cards remained important.

F. Tracking Statistics of Banned Cards

There were two Banlists present within the scope of this project. The statistics of the cards banned on said lists can be graphed.

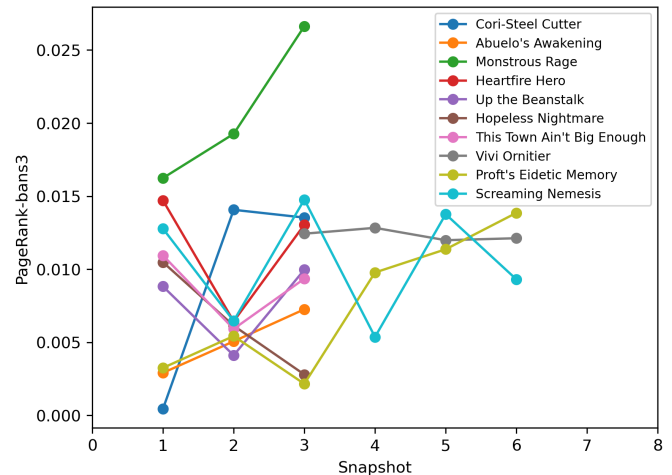


Fig. 4. PageRank - banned cards

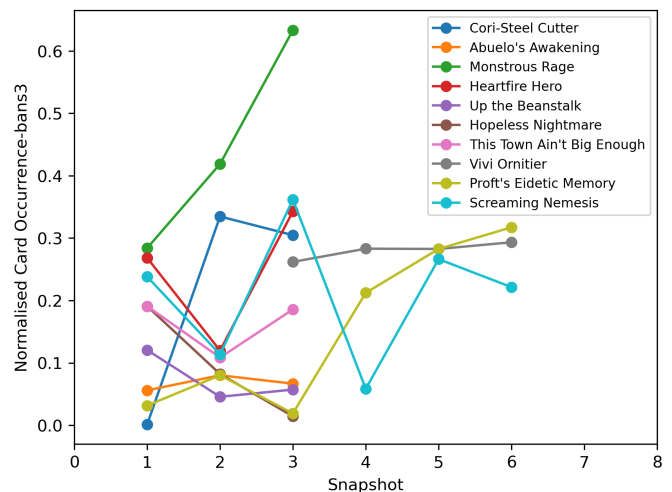


Fig. 5. Normalised card occurrence - banned cards

Fig. 4 and Fig. 5 both detail the PageRank and Normalised card occurrence of the cards that were banned, with the end of their respective lines indicating the snapshot in which they were banned.

For comparison other highly ranked PageRank card's statistics are displayed in Fig. 6 and Fig. 7 to act as a comparison to the cards that were banned. Here banned cards are in red and cards that remained unbanned are in blue.

1-2	Number of cards in community	Com 0	Com 1	Com 2	Com 3	Com 4	Com 5	Com 6	Cards accounted for	Cards unaccounted for
Com 0	59	0.021	0.145	0	0.291	0.007	0.022	0.006	42	17
Com 1	54	0.265	0.047	0	0	0.007	0.011	0.013	38	16
Com 2	49	0.104	0	0.062	0	0	0.09	0	24	25
Com 3	60	0	0	0	0.01	0.28	0	0.019	37	23
Com 4	11	0.02	0.022	0	0	0.085	0	0	11	0
Com 5	11	0	0	0	0.038	0	0	0	2	9
Com 6	10	0	0.022	0.071	0	0	0	0	3	7
Com 7	40	0.157	0.056	0	0	0.008	0	0.157	26	14
Com 8	71	0.032	0	0	0	0.025	0.009	0.336	53	18
Com 9	38	0	0	0	0.013	0	0.254	0.007	17	21

TABLE X

JACCARD SIMILARITY TABLE BETWEEN SNAPSHOTS 1 AND 2

Community	SS2	SS3	SS4	SS5	SS6	SS7	SS8
0	34	11	14	6	24	25	23
1	14	4	21	1	40	32	2
2	14	7	53	21	9	7	14
3	16	7	32	17	42	50	28
4	43	6	10	9	14	14	20
5	10	2	9	19	11	30	12
6	31		14		14	5	
7			50		24	52	
8			40			39	
9			16				
Total	162	37	259	73	165	254	99

TABLE XI

COMMUNITIES NEW CARDS JOINED

SS1 Community 0	SS2 Community 3
"Monstrous Rage" 0.016	"Screaming Nemesis" 0.006
"Emberheart Challenger" 0.015	"Heartfire Hero" 0.006
"Heartfire Hero" 0.015	"Emberheart Challenger" 0.006
"Manifold Mouse" 0.014	"Manifold Mouse" 0.006
"Screaming Nemesis" 0.013	"Hired Claw" 0.005
"Burst Lightning" 0.013	"Witchstalker Frenzy" 0.005
"Torch the Tower" 0.012	"Lightning Strike" 0.003
"Hired Claw" 0.011	"Magebane Lizard" 0.003
"Witchstalker Frenzy" 0.008	"Trumpeting Carnosaur" 0.001
"Pawpatch Recruit" 0.007	"Sunspine Lynx" 0.001

TABLE XII

TOP CARDS BY PAGERANK

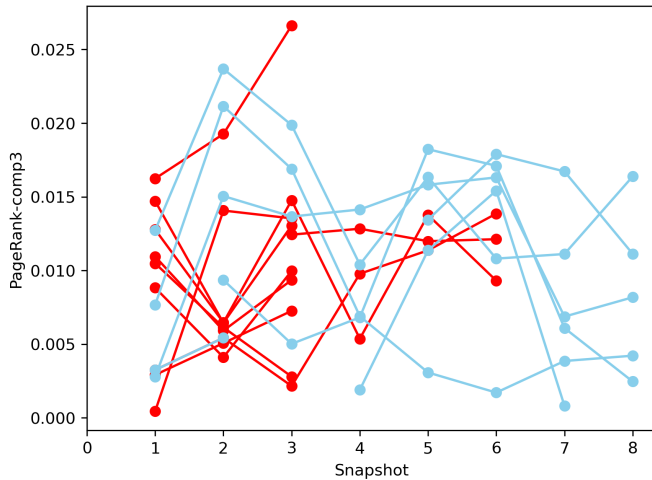


Fig. 6. PageRank - unbanned comparisons

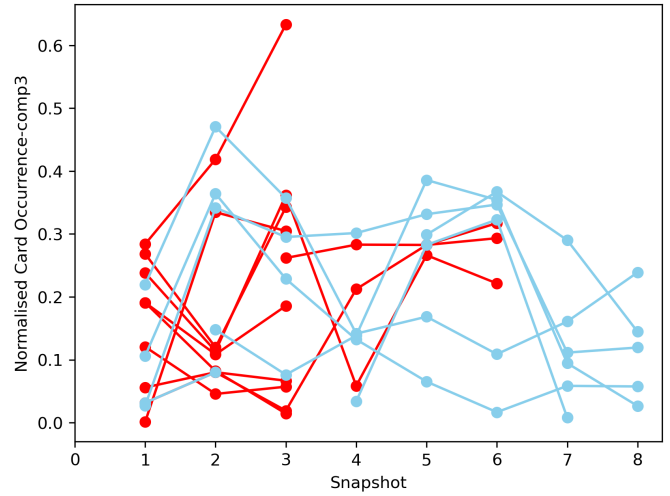


Fig. 7. Normalised card occurrence - unbanned comparisons

within the Graphs. The detected structures were shown to be consistent through the high NMI Scores found within the experiments and clear through the modularity graph.

Detailing the information each graph can show over time can prove to be a useful tool however carelessly taking results as positives or negatives without looking into the reasons behind them can be harmful. The values here can be used as a mathematically backed score for events as number of detected communities will be measurable for each community, however looking further into the events that separate snapshots can give indication as to why a change in diversity may have occurred.

2) *Jumps between Graph Snapshots*: Analysing the jumps between snapshots can also reveal information. Firstly the jumps that saw increases in diversity;

- Between snapshots 3-4 a massive jump in detected communities is seen. Notably snapshot 4 ran for a longer time than snapshot 3 (roughly 2 more weeks), however the results here can still provide useful information. A Banlist was the major event separating snapshots 3-4 and therefore it can be completely justified as it massively increased diversity of the decks seeing tournament play. While both snapshots 3 and 4 have quite high evenness scores snapshot 3 was more balanced than snapshot 4. However that doesn't remove the fact that snapshot 4 still had more communities detected with tournament winning

VIII. DISCUSSION

A. Graph Statistics

1) *Reflection*: It is clear that the consensus Louvain Algorithm can provide stable detection of the communities

results, Its score is dragged down by the 3 communities with only a single tournament topping result each, shown in Table VIII.

- The Jump From 5-6 also saw an increase in diversity when the *Marvel'sSpider – Man* set was introduced. Snapshots 5 and 6 had the highest levels of evenness so the diversity seen through raw community counts is solely where the added diversity comes from. Due to this it could be shown to designers, through these methods that *Marvel'sSpider – Man* was a success for the standard format as it lead to an increase in diversity.
- Finally the Jump From 6-7 increased diversity. This is when the *Avatar : TheLastAirbender* (Avatar) was introduced as well as a Banlist. This points out one of the weaknesses of this analysis' approach, it can be difficult to contribute where specifically diversity stems from if two major events occur. It is known that 60 of Avatar's cards saw immediate competitive success. However these cards along with past ones were most likely given the room to get tournament winning results, through the Banlist which removed the strongest strategies allowing for the reintroduction of diversity.

Next the jumps that lead to decreases in diversity;

- The Jump from snapshot 1-2, seen a large decrease, the major event taking place here was the implementation of the *Tarkir : Dragonstorm* set. This can reveal that the cards of this set were strong and lead to less representation and diversity within the format overall. This is corroborated by the by the 45 cards from *Tarkir : Dragonstorm* that saw immediate competitive success. Unlike with the Avatar set there was no accompanying Banlist that allowed for the introduction of these cards, instead they had to compete with all the current strategies themselves, meaning it was the strength of these cards alone that allowed them to compete at a tournament level. Which along with the raw number that saw play, is further evidence that points towards the new cards being quite strong, which forced past cards out of the meta.
- Snapshot 2-3 also saw a decrease in diversity. This was when the *FinalFantasy* set was introduced, while the communities within this set were quite even there was still a slight drop in diversity from snapshot 2. This drop may mainly stem from the lack of tournament data as snapshot 3's length was one of the shorter time periods taking only 15 days. This points out another flaw with this type of analysis in that the results are only as good as the data and burdened by the lack of it.
- In the jump between snapshots 4-5, a decrease is seen. While *EdgeofEternities* was implemented as a set during this switch, a rotation also took place as well, massively decreasing the available cards at once. This diversity decrease would then make sense as the removal of large number of cards from a format would go onto limit options for players, this is backed up by Table V which had the largest decrease in cards from snapshots 4-5. It is known that 21 of the *EdgeofEternities* cards saw immediate tournament play, however it is clear that

the sweeping change here is a result of Set Rotation despite the implementation of a new set. This makes it difficult to judge *EdgeofEternities* by itself.

- Finally Snapshot 7-8 was another recorded decrease in diversity. Despite the decrease in diversity snapshot 8 was far more even in its distribution of decks than snapshot 7, shown in Fig. 3, so it could be argued that snapshot 8 was more healthy despite 7's higher community count. *LorwynEclipsed* was introduced in this time-frame and 38 of its cards saw immediate competitive success. While it did see a drop in diversity through community count it was also the most even time-frame analysed.

Above are the comments and observations made on the jumps between time-frames, however they can not account for alternative reasons for increases/decreases in diversity. If a set was made primarily with the Commander format in mind it may not implement as well into standard despite being standard legal.

3) *Comparison to User Labels & Diversity of the Snapshot:* NMI ranges from 0.652-0.802, indicating that community membership meaningfully predicts archetype identity despite the number of archetypes vastly outnumbering communities. This trend continues for all snapshots, meaning that if a community is predicted for a deck, its human labelled archetype can be reasonably predicted. Essentially the same groups of archetypes consistently fall into the same communities.

The confidence split is also high for all snapshots meaning that a vast majority of the decks could confidently be placed within a specific community as a majority of their cards lied within it.

The most important scores of this section are however the intra community similarity versus the inter community similarity as this reveals a lot of important information about the clusters. The intra community similarity values range from 0.556 - 0.689 the important information this conveys is that decks within clusters are quite similar. On the other hand the inter community similarity values range from 0.267 - 0.322 this means that decks are quite different from decks within other communities. This confirms that the algorithms used within this project have found real meaning in the structure of their communities as decks within them are similar.

ARI ranges significantly from 0.335-0.773. This is due in large part to the fact that far fewer communities that have been detected than human labels presented. It shows that, despite a large number of decks being labelled as different within human labels they were labelled as the same on the found communities. Looking back to the intra community similarity, decks within those communities can be proven to be similar. The two scores in combination provide very useful information. Despite the fact that these archetypes are labelled differently they were grouped together by community detection and proved to be similar by the intra community scores, this would point towards the realisation that human labels may inflate the diversity of a meta-game by labelling small changes between decks as completely different archetypes. If a number of archetypes with a snapshot are consistently labelled as different by humans and the same by community detection it shows that human labels put a lot of importance

on smaller changes, which are seen to be the formation of a new archetype.

Mentioned earlier was the resolution limit which may affect the modularity maximization based Louvain algorithm, grouping many smaller communities into a singular larger one. Despite this the communities detected were able to show considerable similarity within themselves, suggesting the groupings do have real value and group archetypes that are structurally similar.

The percentage of each deck per snapshot also adds a lot to the diversity analysis. For example snapshot 4 while initially seemed like one of the more diverse snapshots is now less diverse as 3 of its communities have only a single deck within them. This information can help further by more accurately determining how diverse a snapshot is. While communities with very few decks representing them are present, it is important to note that those decks were capable of achieving tournament winning results and their low tournament win rates could be due to any number of reasons not captured by the scope of this project (difficulty to play, etc.).

Both Fig. 3 and Table VIII are vital to providing a more accurate breakdown of the distribution of decks allowing for a more in-depth analysis on the diversity of a format.

4) *Uses*: This temporal analysis aspect of this project can devise a tool which an expert can work alongside with, which can provide mathematical backing to a claim about a format over time. If diversity of a competitive format drops or raises, an expert or designer of the game can look into the major events separating the snapshots and determine if its implementation was strong or weak. In combination with other factors that that expert, designer or player would be aware of, (Player feedback, etc.), they can preform a full analysis of the major event, determining if the process, rules or risks taken for that event should be followed again.

While feedback directly involves a magnitude of players it relies on responses from the community in the forms of surveys to make information manageable. Difficult to manage can also be added to as each player will have different experiences. Using this method as a measure can directly reflect the health of a meta-game without need for human feedback or labels.

This diversity measure aspect of the project may provide an immediate measure of the success of a major event. Which when used in combination with the analysis of an expert can lead to better feedback for game designers and provide players more information for analysis. Both the community count and dispersion of decks within said communities can offer analysts real workable data on how diverse a format is without the need for human labels or the bias that may affect those labels.

As seen by the intra community similarity decks within a community are typically very similar meaning that any two decks within a specific cluster despite being differently labelled archetypes are quite likely to share a large number of cards and in turn playing against both of them can feel very similar. Even if there are a high number of human labelled archetypes within a format if decks still share a large majority of cards they can feel very similar to play against and the

diversity of a format may not feel as high as it is displayed as.

The inter community similarity on the other hand reveals that across communities decks are quite different, this is another important factor as if decks are different across communities it can reinforce the idea that the methods used have found real distinct communities worth using to determine the diversity of a meta-game.

The main advantage this provides is being able to condense similar archetypes within a meta-game, ones that may only differ by a few cards, into a fewer number of communities. This is so that the number of communities may represent the number of vastly different decks and play experiences, better representing the diversity of the game. This measure offers a real way to group up archetypes that are built similarly to allow for an easier analysis of meta-game diversity.

While the diversity of a set can be a difficult measure to both track and explain, a tool devised using the methods of this project can provide assistance to experts. Even though this analysis does seem to have the short coming of difficulty differentiating between multiple major events occurring at the same time, condensing these into a singular event and treating them as such can help designers analysis past events and prepare for future ones.

B. New Card Integration

1) *Reflection*: this method seems initially to be a very fickle measure. As not every card in a set can see tournament success, some cards exist for reasons other than winning tournaments. Another factor of note is cards may drift in and out of any meta-game, making the tracking of the total unique cards that see any tournament play difficult to measure. It is possible for a card to be introduced in snapshot 1, not win at all during snapshot 2 (and by extension not show up in the graph), then reappear on the graph of snapshot 3.

Another issue is a card may only rejoin the graph due to future cards release meaning that feedback for the original set would be difficult to give, given that that future card had not existed at the time.

2) *Uses*: while this alone is not recommended as a measure of success for a set, as there can be many aspects to how “good” a set is, (fun, alternative formats, etc.), this can still provide immediate feedback to designers signalling if a set was too strong initially, (a very high percentage of new cards saw play), or if its power-level was inline with past sets. For example *Avatar : TheLastAirbender* can be seen as stronger than the other sets as 60 of its cards saw immediate competitive success.

C. Tracking Communities Overtime

1) *Reflection*: in the results section, two examples of tracking communities were presented one where snapshot 1 community 0 became snapshot 2 community 3 and another where snapshot 1 community 5 died out. This process can be iteratively applied to snapshots to determine when a community dies out and for how long it lived.

Additional examples of tracking communities over time to prove it’s effectiveness are available through Appendix B.

2) *Uses*: while sets are not created immediately before release this tracking of communities over time can be employed to track specific archetypes. Since new sets are made months in advance this information can be used to show how long archetypes have lived and assuming they'll live until the current set in development has been released, this can allow designers notice ahead of time to keep this archetype in mind for balancing and gameplay choices.

D. Tracking Banned Cards Overtime

1) *Reflection*: Here specifically the information carried by the snapshot's graphs, both PageRanking's of cards and the normalised card occurrences, were used to see if banned cards had consistent statistics which could be used to argue for the banning of future cards.

Unfortunately both the banned cards and some of the high ranked PageRank cards of the time have quite similar scores with some of the cards that didn't get banned scoring higher PageRanks and usage rates than some of the banned cards. On the comparison graphs Fig. 6 and Fig. 7 the banned cards are very difficult to distinguish from the ones that remained unbanned, and therefore difficult to find trends in the cards for the two aforementioned statistics.

This does mean that the PageRank and normalised card occurrences derived from this graph are incapable of providing evidence that supports if a card should be banned.

This however can be in large part due to the volatile nature of CCGs. With cards constantly releasing cards can be powerful in one format and weak in the next.

Finally cards may be banned for reasons other than power. If a card is banned it may be due to other number of reasons outside the scope of this project, (a card design that does not promote fun play, etc.)

IX. CONCLUSION

A. Research Questions Answered

1) Q: Is an unlabelled clustering approach a viable way of assessing the current state of a format and by extension a viable way of performing a temporal analysis?

A: From this approach human labelled archetypes can be successfully condensed into consistently detectable communities. They differ from human labels, however have proven to show clear structure, grouping similar archetypes and within these communities decks show clear similarity and across which they show clear difference. As this information can confidently produce graphs, it is viable for a temporal analysis.

2) Q: Can the data provided by a temporal analysis this way provide feedback to designers free from human bias?

A: Instead of small differences in decks inflating a time periods diversity, detected communities can be used as a bias-less measure of diversity, which overtime can be used as comparisons of healthy formats to unhealthy ones and can act as targets for designers to aim towards. These communities can be tracked and measured on evenness and in combination with community counts can reveal how diverse a time-frame within a format truly

is. Communities can be tracked over time providing an expert an idea of how long they lived and at what point they died or got absorbed by another community.

3) Can this Framework be applied to other formats and CCGs?

This framework is fast and has no reliance on anything specific to MTG or the Standard MTG format. Meaning this framework could be adapted as an unsupervised approach that could be used to analyse an alternative format or CCG.

B. Future work

1) *Graph overviews*: This aspect of the project could be expanded upon so that it covers a larger time frame. The limited selected time frame for this project was a product of the available time to work on this project, however with this system, built up over time and running alongside the present, would be able to consistently inform designers how the choices they made have affected the game and for future set releases detail good methodologies to follow. If a set preformed well, was received well by players and mathematically shows an increase in diversity, that set's direction, methodology, rules while designing and more can be applied to future set creation.

2) *New Card Integration*: This aspect of the project could be expanded upon so that over the course of every snapshot the portion of its cards that stem from a specific set is tracked, this could add more of a long term feedback for designers but would provide little immediate feedback.

3) *Community detection algorithms*: different Community detection algorithms and experimenting with alternative Louvain parameters could be added to to see if smaller communities could be produced aligning closer to human made labels.

4) *Framework application*: This framework could be applied to another MTG format and seeing if the varying power levels and expected performance within those formats see different outputs and uses for this type of analysis. Applying this framework to alternate CCGs could also be done and seeing if their alternative structures make analysis through this method easier or more difficult.

APPENDIX A USAGE OF AI

AI (Chat GPT) was used within this project in several ways;

- 1) AI was used to help with testing small sections of code ensuring that it had run correctly, (e.g. AI helped find the cause of an infinite loop, which had been caused by an incorrect setup of stopping conditions).
- 2) AI was used to help get a timeline of various MTG events, ensuring snapshots could be separated into the correct time-frames. These dates were all verified by the author.
- 3) AI was used to help the author learn about Overleaf, how to use the commands correctly and how to correctly format citing within it. (e.g. AI helped format the references below).
- 4) AI was used to help the author determine what the typical (network science) graphical analysis pipeline looked like, so that it could be adapted to this project.

- 5) Finally AI was used to help the author understand how some concepts worked on a fundamental level. (Such as why natural log is used in the Shannon function).

APPENDIX B

ADDITIONAL EXAMPLES OF TRACKING COMMUNITIES OVER TIME

The additional examples for the tracking communities over time through the use of Jaccard tables and the PageRanking of important cards is available as a PDF on this project's GitHub page: <https://github.com/Liam-Caffrey/Capstone-Project>

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REFERENCES

- [1] G. N. Yannakakis and J. Togelius, *Artificial Intelligence and Games*, 2nd ed. Springer Nature, 2025. <https://gameaibook.org>
- [2] C. Alvin, M. Bowling, S. Rivers-Green, D. Siglin, and L. Alvin, "Toward a Competitive Agent Framework for Magic: The Gathering," *The International FLAIRS Conference Proceedings*, vol.34, 2021.
- [3] S. Fortunato and D. Hric, "Community detection in networks: A user guide," *Physics Reports*, vol. 659, pp. 1–44, 2016.
- [4] M. E. J. Newman, "Modularity and community structure in networks" *Proc. Natl. Acad. Sci.*, vol. 103, no.23, pp. 8577–8582, 2006.
- [5] S. Fortunato and M. Barthélémy, "Resolution limit in community detection," *Proc. Natl. Acad. Sci.*, vol. 104, no. 1, pp. 36–41, 2007.
- [6] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *J. Stat. Mech. Theory Exp.*, vol. 2008, no. 10, p. P10008, 2008.
- [7] A. Lancichinetti and S. Fortunato, "Consensus clustering in complex networks," *Scientific Reports*, vol. 2, Art. no. 336, pp. 1–7, 2012.
- [8] L. Danon, A. Díaz-Guilera, J. Duch, and A. Arenas, "comparing community structure identification" *J. Stat. Mech. Theory Exp.*, no. 9, pp. 219–228, 2005.
- [9] L. Hubert and P. Arabie, "Comparing partitions," *J. Classif.*, vol. 2, no. 1, pp. 193–218, 1985.
- [10] C. E. Shannon, "A mathematical theory of communication," *Bell Syst. Tech. J.*, vol. 27, no. 3, pp. 379–423, 1948.
- [11] L. Jost, "The relation between evenness and diversity," *Diversity*, vol. 2, no. 2, pp. 207–232, 2010.
- [12] S. Brin and L. Page, "The anatomy of a large-scale hypertextual Web search engine," *Comput. Networks ISDN Syst.*, vol. 30, pp. 107–117, 1998.
- [13] MTGTop8, *Magic: The Gathering Decklists Database*, [Online]. Available: <https://mtgtop8.com/index>
- [14] Scryfall, *Oracle Cards Bulk Data* dataset version oracle-cards-20260508090243, May 8, 2026. <https://scryfall.com/docs/api/bulk-data>